

ACmK-OT Clinical Operations Platform

This document defines the formal system architecture, backend state machines, crisis coordination logic, audit/compliance framework, FHIR interoperability strategy, and Board-of-Nursing alignment for the ACmK-OT system.

1. RoleSession & Assignment Engine (C++)

The RoleSession engine governs authentication, authority, and shift accountability. Assignments bind nurses to units, floors, rooms, and patients with deterministic reassignment rules.

2. Crisis Coordinator

The Crisis Coordinator manages internal and inbound emergencies, staffing acknowledgements, temporary patient redistribution, and safe restoration of assignments post-crisis.

3. Audit & Compliance Ledger

Every clinical action is recorded as an immutable ledger entry including actor, role, time, patient, action, and outcome to meet legal, regulatory, and forensic requirements.

4. FHIR Adapter Layer

The system exposes optional FHIR R4-compatible resources through a translation adapter, allowing interoperability without compromising native performance or sovereignty.

5. Board of Nursing Compliance

This platform enforces scope-of-practice boundaries, individual accountability, real-time supervision, and documentation integrity consistent with U.S. State Boards of Nursing requirements.